



ReliefTrack PH: Community Disaster and Aid Management System

Brgy. San Perfecto ReliefTrack

SDG 9 - Industry, Innovation and Infrastructure

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Introduction

The ReliefTrack PH is designed to improve how disaster relief operations are managed in local communities, particularly in Barangay San Perfecto. During emergencies, organizing information, distributing resources, and coordinating people can become challenging. This system aims to address those challenges by providing a centralized and organized digital platform.

ReliefTrack PH supports key processes such as resident registration, relief request handling, donation tracking, and task management for volunteers and officials. By using this system, barangay officials can make faster and more accurate decisions, while residents can easily request assistance when needed. It also promotes transparency and accountability by keeping clear records of all activities.

ReliefTrack PH is a role-based disaster management system designed for barangay-level operations in the Philippines. Unlike generic admin-based systems, ReliefTrack PH distributes administrative authority among elected barangay officials (Barangay Chairman and SK Chairman), reflecting actual LGU hierarchy.

The system supports five primary actors:

1. Resident
2. SK Kagawad
3. SK Chairman
4. Barangay Chairman
5. Donor



I. Project Charter

Objectives

1. To develop a role-based digital platform for barangay disaster management in Brgy. San Perfecto.
2. To automate the relief request process from submission to distribution tracking.
3. To provide real-time visibility of relief operations to all stakeholders (Residents, SK Kagawad, SK Chairman, Barangay Chairman, Donors).
4. To ensure accountability through immutable audit logging of all system actions.
5. To replace manual paper-based processes with an efficient digital infrastructure aligned with SDG 9.

SMART Success Criteria

Criteria	Description
Specific	The system must support 5 user roles with distinct dashboards and enable end-to-end relief request processing.
Measurable	100% of relief requests must be processed within 48 hours; audit logs must capture 100% of user actions; report generation must complete in <10 seconds.
Achievable	Using standard web technologies (HTML, CSS, JavaScript) with client-side data management
Relevant	Directly addresses SDG 9 (Industry, Innovation, Infrastructure) by digitizing barangay disaster operations.
Time-bound	Core system delivery within 8 weeks; full deployment within 10 weeks.



High-level Risks

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R-01	Low digital literacy among barangay residents	High	Medium	Provide simple UI with guided tutorials and demo accounts
R-02	Intermittent internet connectivity during disasters	Medium	High	Implement offline-capable forms with sync functionality
R-03	Resistance to change from manual to digital system	Medium	Low	Conduct training sessions for barangay officials
R-04	Data privacy concerns under RA 10173	Low	High	Implement audit trails and role-based access control
R-05	System downtime during peak disaster declaration	Low	Medium	Ensure 99.5% uptime with lightweight architecture



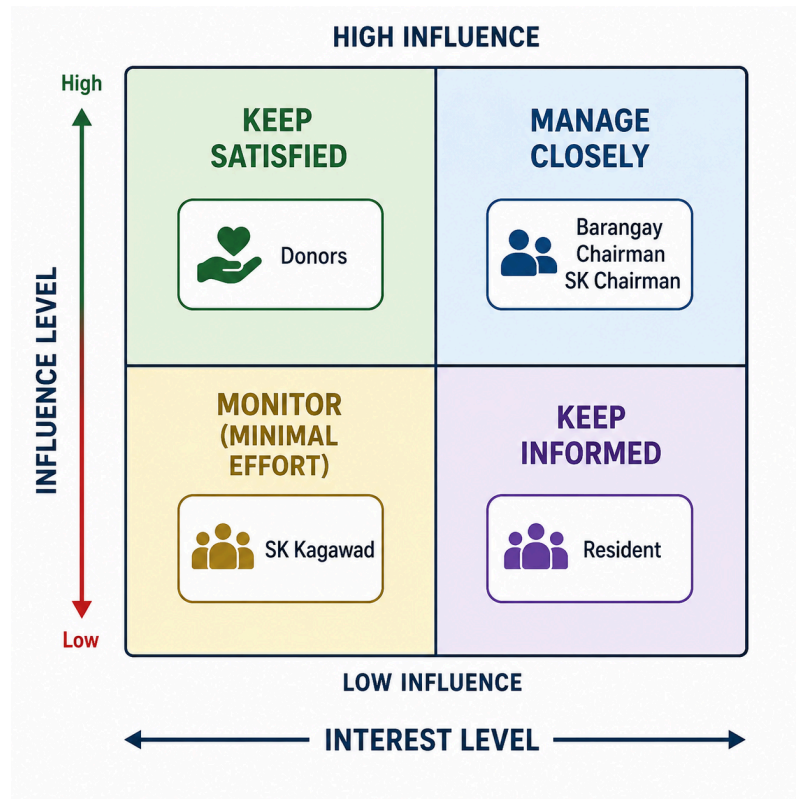
II. Stakeholder Analysis

Stakeholder Register

Stakeholder	Role in System	Interest	Influence	Power
Barangay Chairman	Final approver, validator	High	High	Decision maker
SK Chairman	Operations planner, task assigner	High	Medium	Operational lead
Residents	Request submitters, status checkers	High	Low	End users
Donors	Donation submitters	Medium	Low	Funding source
SK Kagawad	Field implementer	Medium	Low	Frontline workers



Interest/Influence Matrix



Negotiation Log

Issue	Stakeholder(s) Involved	Resolution	Status
Manual approval process too slow	Barangay Chairman, Resident	Implemented digital approval with real-time notifications	Resolved
SK Kagawad needs mobile access	SK Kagawad, SK Chairman	Developed responsive design for mobile devices	Resolved
Donors want transparency reports	Donors, Barangay Chairman	Added Transparency Report dashboard for donors	Resolved



Residents concerned about data privacy	All users	Implemented audit logs and RBAC per RA 10173	Resolved
Delegation of authority needed	Barangay Chairman	Added Delegate Authority feature for temporary transfer	Resolved

III. Requirements Elicitation

Interview Protocol

Interview Title: ReliefTrack PH Requirements Gathering

Date: May 2026

Interviewer: System Analysis Team

Interviewees: Barangay Chairman, SK Chairman, SK Kagawad, Residents, Donors

Questions Asked:

1. "What are the biggest challenges in your current disaster relief process?"
2. "What information do you need to see immediately when you log into the system?"
3. "What manual steps would you like to see automated?"
4. "Who should have authority to approve relief requests?"
5. "How do you currently track task assignments to volunteers?"
6. "What transparency information would donors want to see?"

Raw Notes Summary

Interviewee Role	Key Feedback
Barangay Chairman	"Verbal approvals cause delays. A digital approval system with an audit trail is needed."
SK Chairman	"We need to assign tasks to SK Kagawad members and monitor their progress in real time."
SK Kagawad	"Mobile accessibility is important because we are often deployed in the field."



Resident “We want to know the status of our requests without going to the barangay hall.”

Donor “We want transparency regarding how donations are being utilized.”

IV. Specific Requirements

Functional Requirements (MoSCoW Prioritization)

MUST HAVE

ID	Requirement
FR-01.1	The system shall allow residents to register using their name, address, and contact number.
FR-01.2	The system shall allow residents to upload a valid government ID for identity verification.
FR-01.3	The system shall allow residents to submit relief requests specifying the type and quantity of goods needed.
FR-01.4	The system shall allow residents to view the status of submitted relief requests.
FR-02.1	The system shall allow SK Kagawad to view tasks assigned by the SK Chairman.



FR-03.1	The system shall allow the SK Chairman to create and plan relief operations.
FR-03.2	The system shall allow the SK Chairman to assign tasks to SK Kagawad members.
FR-04.1	The system shall allow the Barangay Chairman to review and approve or deny resident relief requests.
FR-04.2	The system shall allow the Barangay Chairman to validate resident information and uploaded IDs.
FR-04.5	The system shall allow the Barangay Chairman to view system audit logs.
FR-05.1	The system shall allow donors to submit donation details including type, quantity, and date.
FR-06.1	The system shall require secure login and logout for all users.
FR-06.2	The system shall secure user passwords before storage.
FR-06.3	The system shall automatically log out users after 15 minutes of inactivity.



FR-06.4	The system shall enforce role-based access control (RBAC) for all features.
FR-06.5	The system shall log all user actions with timestamps and user IDs
FR-06.6	The system shall prevent any action from an unauthenticated or unauthorized user.

SHOULD HAVE

ID	Requirement
FR-01.5	The system should allow residents to view the distribution status of approved requests.
FR-02.4	The system should allow SK Kagawad to flag operational issues.
FR-03.5	The system should allow the SK Chairman to view operational inventory levels
FR-04.6	The system should allow the Barangay Chairman to delegate approval authority temporarily.
FR-05.3	The system should allow donors to view transparency reports.



COULD HAVE

ID	Requirement
FR-02.3	The system could allow SK Kagawad to record field activity updates.
FR-03.4	The system could allow the SK Chairman to oversee distribution operations.
FR-04.4	The system could allow the Barangay Chairman to view full inventory reports.

WON'T HAVE

ID	Requirement	Reason For Exclusion
FR-W01	Advanced data encryption (AES-256 for stored data)	Excluded from current version due to project scope; will be implemented in future deployment
FR-W02	Biometric authentication	Infrastructure not available in target barangay
FR-W03	SMS notifications	Requires third-party gateway; email sufficient for v1
FR-W04	Native mobile apps (iOS/Android)	PWA sufficient for v1; lower development cost



Non-Functional Requirements

ID	Category	Requirement	Priority
NFR-01	Performance	The system shall generate reports in less than 10 seconds	MUST
NFR-02	Security	The system shall have no generic Admin role; all sensitive actions require role verification	MUST
NFR-03	Auditing	The system shall log all CRUD operations involving requests, donations, and approvals	MUST
NFR-04	Usability	The system shall display only role-relevant dashboards after login	MUST
NFR-05	Availability	The system should remain accessible during peak disaster operations with minimal downtime	SHOULD

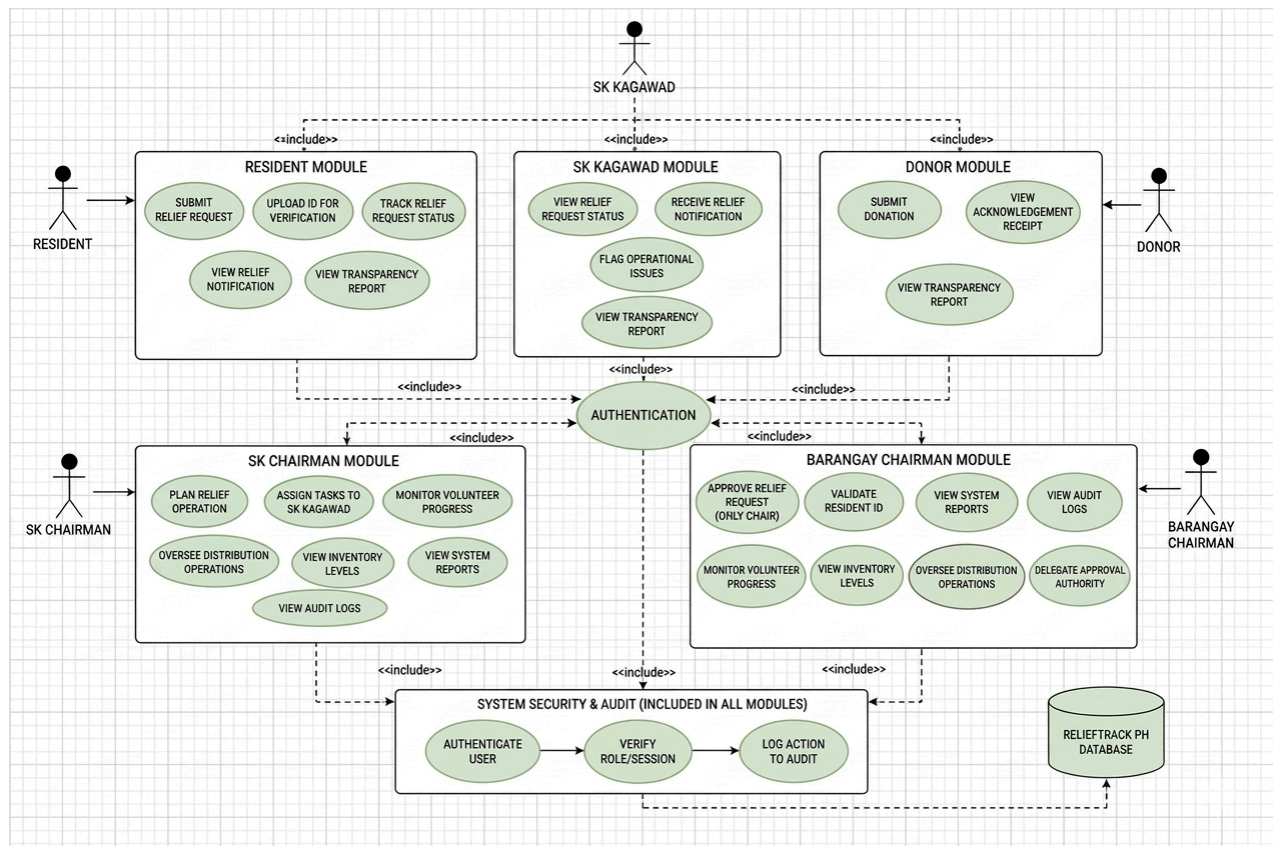


NFR-06	Delegation	The system shall automatically expire temporary authority transfers after a specified duration	SHOULD
NFR-07	Compatibility	The system shall support modern web browsers (Chrome, Firefox, Safari, Edge)	SHOULD
NFR-08	Maintainability	The system shall allow easy updating and maintenance of records and modules	SHOULD
NFR-09	Performance	Page load time on 4G network shall be less than 2 seconds	MUST
NFR-10	Security	Failed login attempts shall be limited to 5 attempts per 15 minutes	SHOULD



IV. Logical Use Case Models

The Use Case Diagram visually represents the interaction between the system actors and the major functions of ReliefTrack PH. The system supports five primary actors: Resident, SK Kagawad, SK Chairman, Barangay Chairman, and Donor.



This illustrates a role-based, five-module ecosystem, sort of centered around a central Authentication hub and a bottom-tier System Security & Audit layer. Inside the boxes, the Resident module and the Donor module handle citizen-facing things like submitting relief requests, tracking status, getting notifications, and logging donations. Meanwhile the SK Kagawad module, SK Chairman module, and Barangay Chairman module are for internal administrative use cases that range from monitoring inventory to tracking task statuses, and yes also enforcing a strict centralized request approval power but exclusively under the Barangay Chairman.



Use Case 1: Submit Relief Request (UC-01)

Element	Description
Use Case ID	UC-01
Use Case Name	Submit Relief Request
Actor	Resident
Description	Residents submit a request for relief goods to the barangay.
Precondition	Residents are logged into the system.
Postcondition	The request is saved with status "PENDING" and the Barangay Chairman is notified.
Main Flow	1. Residents click the "New Relief Request" button. 2. System displays request form. 3. Residents fill in items, quantity, and reason. 4. The resident clicks "Submit Request". 5. System validates input. 6. The system saves requests with the status "PENDING". 7. System logs action to audit trail. 8. System confirms submission to residents.
Alternative Flow	4a. Resident cancels form: System returns to dashboard. 5a. Missing required fields: System displays error message.
Exception Flow	6a. Database error: System displays error and prompts retry.
Business Rules	BR-01: Only verified residents can submit requests.



Use Case 2: Approve Relief Request (UC-02)

Element	Description
Use Case ID	UC-02
Use Case Name	Approve Relief Request
Actor	Barangay Chairman
Description	The Barangay Chairman reviews and approves a pending relief request.
Precondition	The chairman is logged in. Pending requests exist.
Postcondition	Request status is updated. Residents are notified.
Main Flow	1. Barangay Chairman clicks "Pending Approvals" 2. System displays list of pending requests 3. Chairman selects a request to review 4. System displays request details (items, quantity, reason) 5. Chairman reviews resident information and uploaded ID 6. Chairman clicks "Approve" button 7. System asks for confirmation 8. Chairman confirms 9. System updates status to "APPROVED" 10. System logs approval to audit trail 11. System notifies resident of approval
Alternative Flow	6a. Chairman clicks "Reject": System asks for rejection reason, saves as "REJECTED".
Exception Flow	9a. Update fails: System rolls back and displays error.
Business Rules	BR-03: Only Barangay Chairman can approve requests.



	BR-04: Approval requires residents to be verified.
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Use Case 3: Assign Task to SK Kagawad (UC-03)

Element	Description
Use Case ID	UC-03
Use Case Name	Assign Task to SK Kagawad
Actor	SK Chairman
Description	SK Chairman assigns a distribution task to an SK Kagawad member.
Precondition	SK Chairman is logged in. A relief operation is planned.
Postcondition	The task is saved and visible to assigned SK Kagawad.
Main Flow	1. SK Chairman clicks "Assign Tasks". 2. System displays assignment form. 3. SK Chairman selects SK Kagawad from the dropdown. 4. SK Chairman enters task description and due date. 5. SK Chairman clicks "Assign Task". 6. System validates input. 7. The system saves tasks with the status "PENDING". 8. System logs assignment to audit trail. 9. System confirms assignment.
Alternative Flow	4a. SK Chairman leaves due date empty: System defaults to 7 days from today.
Business Ruls	BR-05: Task must have a due date within 30 days.



	BR-06: Task can only be assigned to users with role "SK Kagawad".
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VI. System Boundary

Context Diagram (Level -1 DFD)

